

Monday Warm Up: Unit 1, Capacitors, current and DC circuits

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1 Memory Bank

- $\vec{E}_{\text{plane}} = \frac{\sigma}{2\epsilon_0}\hat{k} = \frac{Q}{2A\epsilon_0}\hat{k}$... The electric field of a plate or disk with charge Q and area A , for distances much smaller than the radius or length.
- $\Delta U = q\Delta V$... Relationship between change in potential energy, charge, and voltage.
- $V(r) = kq/r$... Potential (voltage) of a point charge.
- $V(x) = Ex + V_0$... Linear relationship between voltage, electric field, and distance between two charge plates to form a capacitor.
- $Q = C\Delta V$... Relationship between stored charge, capacitance, and voltage.
- $I = \Delta Q/\Delta t$... The relationship between current, I , and charge.
- $C_{\text{tot}}^{-1} = C_1^{-1} + C_2^{-1}$... Two capacitors *in series*.
- $C_{\text{tot}} = C_1 + C_2$... Two capacitors *in parallel*.
- $V_C(t) = \mathcal{E}(1 - \exp(-t/\tau))$... The capacitor voltage in an RC circuit, with $\tau = RC$.

2 Review: Electric Fields and Charge

1. Consider a particle of charge q and mass m passing through the region between a pair of infinitely long horizontal plates separated by a distance d with a uniform electric field strength E in the $-\hat{y}$ direction. At $t = 0$, the particle is at the origin, half-way between the plates, moving horizontally with velocity v . (a) Develop an expression for the horizontal displacement, Δx , the particle has when it strikes one of the plates. (b) Calculate Δx , Assuming $E = 500 \text{ V m}^{-1}$, $m = 7 \times 10^{-27} \text{ kg}$, $q = 3.2 \times 10^{-19} \text{ C}$, $d = 3 \text{ cm}$, and $v = 300 \text{ m s}^{-1}$.

3 Capacitors and Current

1. Current has a unit of 1 Amp (A), or 1 C s^{-1} . (a) Suppose we allow 5 mA of current to flow onto a capacitor for 5 seconds. How much charge is stored in the capacitor? (b) After 5 seconds, the capacitor voltage is measured to be 5.0 V. What is the capacitance?
2. What capacitance is needed to store $3.00 \mu\text{C}$ of charge at a voltage of 120 V?
3. Suppose an RC circuit, with $\mathcal{E} = 5.0 \text{ Volts}$, has $R = 1 \text{ k}\Omega$, and $C = 1 \mu\text{C}$. (a) What is $\tau = RC$? in ms? (b) What is the voltage in the capacitor, V_C , 10 ms after the circuit is connected?
4. (a) Two $1 \mu\text{F}$ capacitors are connected *in series*. What is the total capacitance? (b) Two $1 \mu\text{F}$ capacitors are connected *in parallel*. What is the total capacitance?